higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS

(First Paper)

NQF LEVEL 3

(10501053)

20 February 2019 (X-Paper)
09:00 –12:00

Nonprogrammable scientific calculators may be used

This question paper consists of 7 pages, 1 formula sheet and 2 addenda.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Clearly show ALL calculations, diagrams, graphs, et cetera which you have used in determining the answers.
5. If necessary, answers should be rounded off to THREE decimal places, unless stated otherwise.
6. Diagrams are NOT drawn to scale.
7. Write the EXAMINATION NUMBER and CENTRE NUMBER in the spaces provided on ADDENDUM A and B (attached). Detach them from the question paper and hand them in with the ANSWER BOOK.
8. Write neatly and legibly.

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QUESTION 1

1.1 Without the use of a calculator and without converting to polar form, simplify the following complex numbers to the standard form of $a+bj$.

1.1.1 $(5-2i)^2$ (3)

1.1.2 $(12+4i)-i^3(6-3i)$ (4)

1.1.3 $7i^3-2-3i-8i^2+\sqrt{16}$ (3)

1.1.4 $\dfrac{\sqrt{-16}-5}{3+5i}$ (5)

1.2 Simplify to the polar form, $rcis\theta$:

1.2.1 $\dfrac{\sqrt{3}}{2}-\dfrac{1}{2}i$ (4)

1.2.2 $2cis20{,}2°\times 30cis60{,}8°$ (3)

1.3 Simplify the following to the rectangular form of $a+bj$:

$3cis133°+2cis20°$ (5)

1.4 Given $z=-7i-1$:

1.4.1 Write down the conjugate of z. (1)

1.4.2 Calculate the argument (θ) of $\bar{z}$. (2)

[30]

QUESTION 2

2.1 Use differentiation rules to determine $\dfrac{dy}{dx}$ of the following: (Leave your answer with a POSITIVE exponent and in SURD form where applicable.)

2.1.1 $y=2x^3(2x+7)$ (3)

2.1.2 $y=\dfrac{\sqrt{x}+x-\sqrt[3]{x}}{x^3}$ (3)

2.1.3 $y=\dfrac{x^3}{3}-\sqrt{x}-8x$ (3)

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2.2 Evaluate: $\lim_{x\to -1}\frac{x^2+x}{x+1}$ (2)

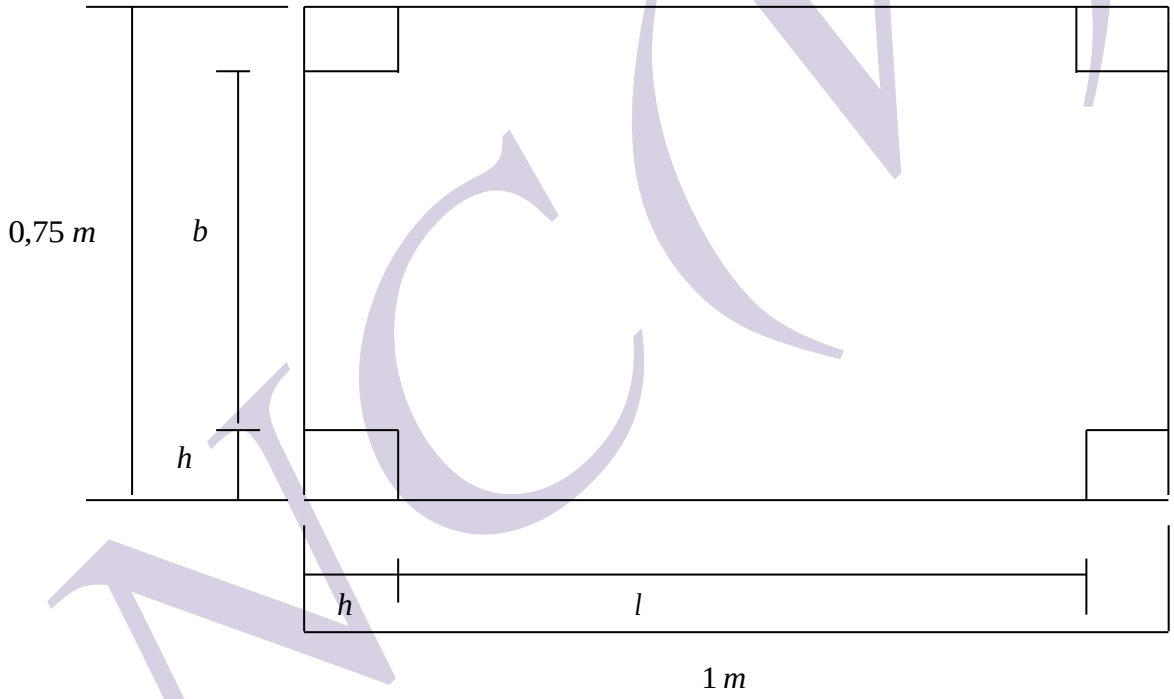
2.3 Determine $f'(x)$ from the first principles if $f(x)=2-x^2$. (4)

2.4 A cardboard box without a lid must be constructed from a sheet of cardboard of 1 m × 0,75 m.

What should the dimensions of the box be to contain the maximum volume possible? Square corners should be cut out of the cardboard sheet to form the sides.

HINT: $v = l \times b \times h$

NOTE: The diagram is not drawn to scale.



(5)
[20]

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QUESTION 3

3.1 Simplify the following expressions:

3.1.1 $$\frac{x^2-1}{x^3-2x^2-x+2} \div \frac{3x+6}{x^2-4}$$ (4)

3.1.2 $$\frac{x}{x+y}+\frac{y}{x}$$ (3)

3.2 Solve for x by using a quadratic formula:

$$3x^2-8+2x=-x^2+4$$ (4)

3.3 Solve for x:

$$x^2-3x-13<-9$$ (4)

3.4 Solve for x and y:

$$y=x^2-3x$$
$$y-x=-4$$ (5)

3.5 A factory produces a number of articles per week. The following system of inequalities were obtained:

$$50 \le x \le 200$$
$$y \ge 100$$
$$x+y \le 400$$
$$x+y \ge 200$$

3.5.1 Use the graph paper on ADDENDUM A (attached) to draw the graphs with the given constraints.

Write the EXAMINATION NUMBER and CENTRE NUMBER in the spaces provided on ADDENDUM A, detach it from the question paper and hand it in with the ANSWER BOOK. (5)

3.5.2 Determine the feasible region. (1)

3.5.3 Find the values of x and y which will maximise $P=30x+10y+8\,000$. (4)

[30]

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QUESTION 4

4.1 Given the functions $f(x) = -x^2 - 2x + 8$ and $g(x) = 2^{-x} + 2$:

4.1.1 Use the graph paper on ADDENDUM B (attached) to draw the graphs on the same set of axes.

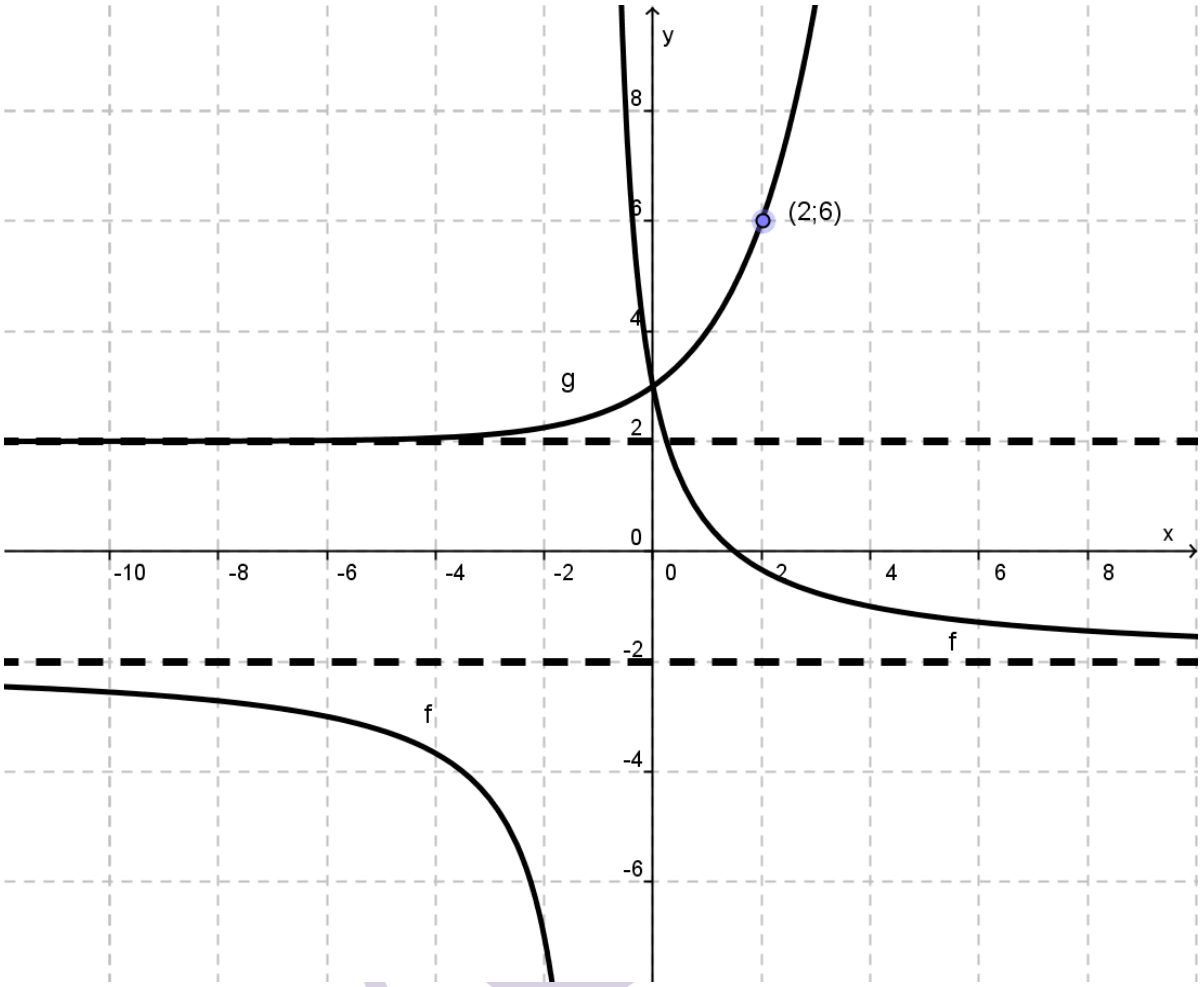
Write the EXAMINATION NUMBER and CENTRE NUMBER in the spaces provided on ADDENDUM B, detach it from the question paper and hand it in with the ANSWER BOOK. (8)

4.1.2 Write down the asymptote of $g(x)$. (1)

4.1.3 Determine the domain of $f(x)$. (1)

NCW

4.2 The following are graphs of $f(x)=\frac{a}{x+p}+q$ and $g(x)=b^x+c$.



4.2.1 For which value(s) of x is:

$$\frac{a}{x+p}+q=b^x+c$$ (1)

4.2.2 Write down the coordinates of the x- and y-intercept of f. (2)

4.2.3 What are the equations of the asymptotes of f? (2)

4.2.4 Determine the equation of f. (4)

4.2.5 Write down the equations of the asymptotes of g. (1)

[20]

TOTAL: 100

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FORMULA SHEET

1. $z = r\cos\theta + r\ j\sin\theta$

2. $z = a \pm bj$ or $z = a \pm bi$ *where* $i = j = \sqrt{-1}$

3. $r = \sqrt{a^2 + b^2}$ or $r = \sqrt{z \times \bar{z}}$

4. $\alpha = \tan^{-1}\left(\frac{b}{a}\right)$

5. $r \angle\theta = r\ cis\ \theta$

6. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

7. $y = ax^2 + bx + c$

8. $y = a(x - p)^2 + q$

9. $y = a(x - x_1)(x - x_2)$

10. $y = \frac{a}{(x + p)} + q$

11. $f'(x) = \lim_{h \to 0} \frac{f(x + h) - f(x)}{h}$

12. $\frac{d}{dx} x^n = nx^{n-1}$

13. $\frac{d}{dx} k = 0$

14. $Dx[kf(x)] = kDx[f(x)]$

15. $Dx[f(x) \pm g(x)] = Dx[f(x)] \pm Dx[g(x)]$

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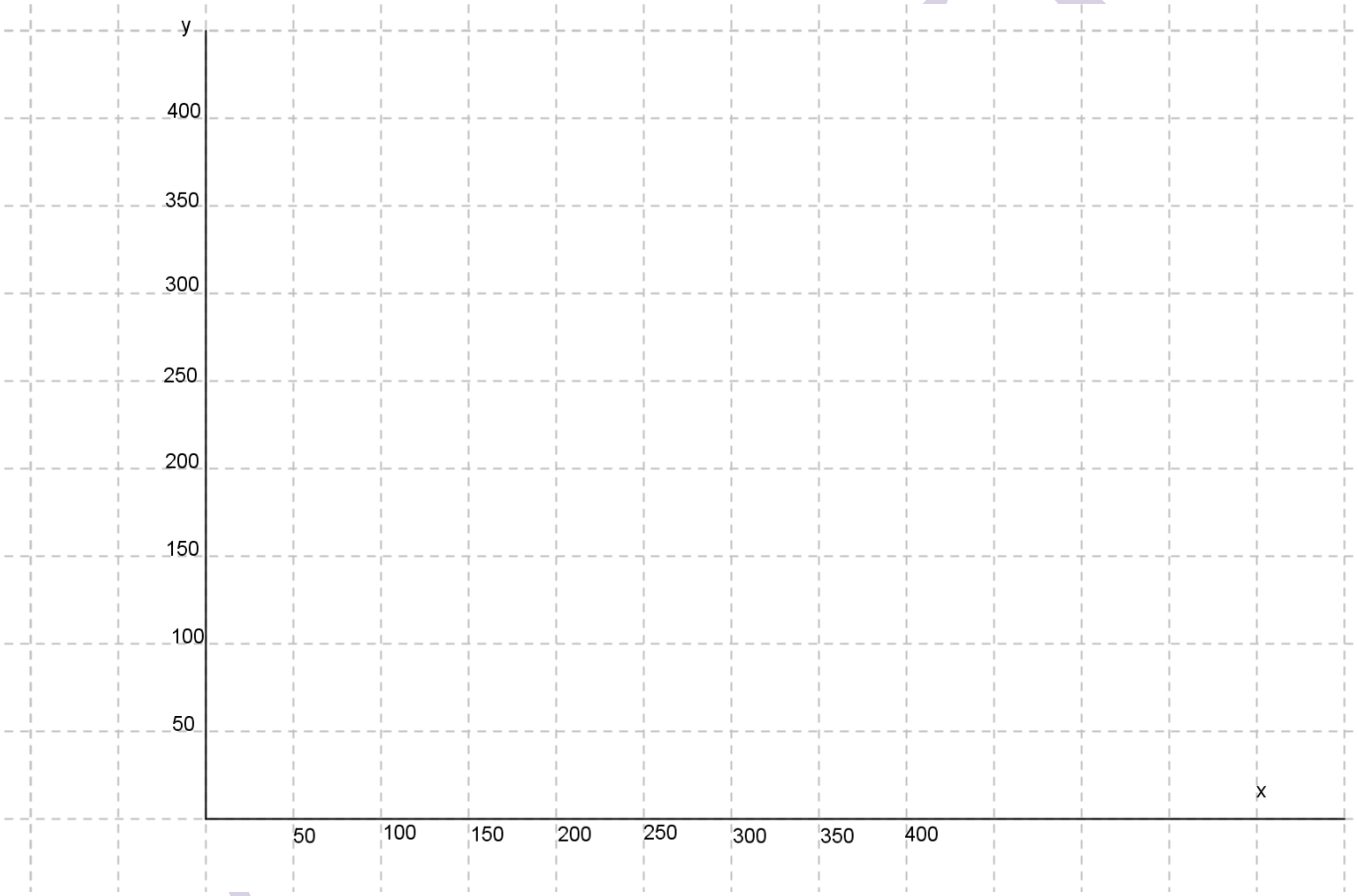
ADDENDUM A

EXAMINATION NUMBER:

CENTRE NUMBER:

QUESTION 3

3.5 3.5.1



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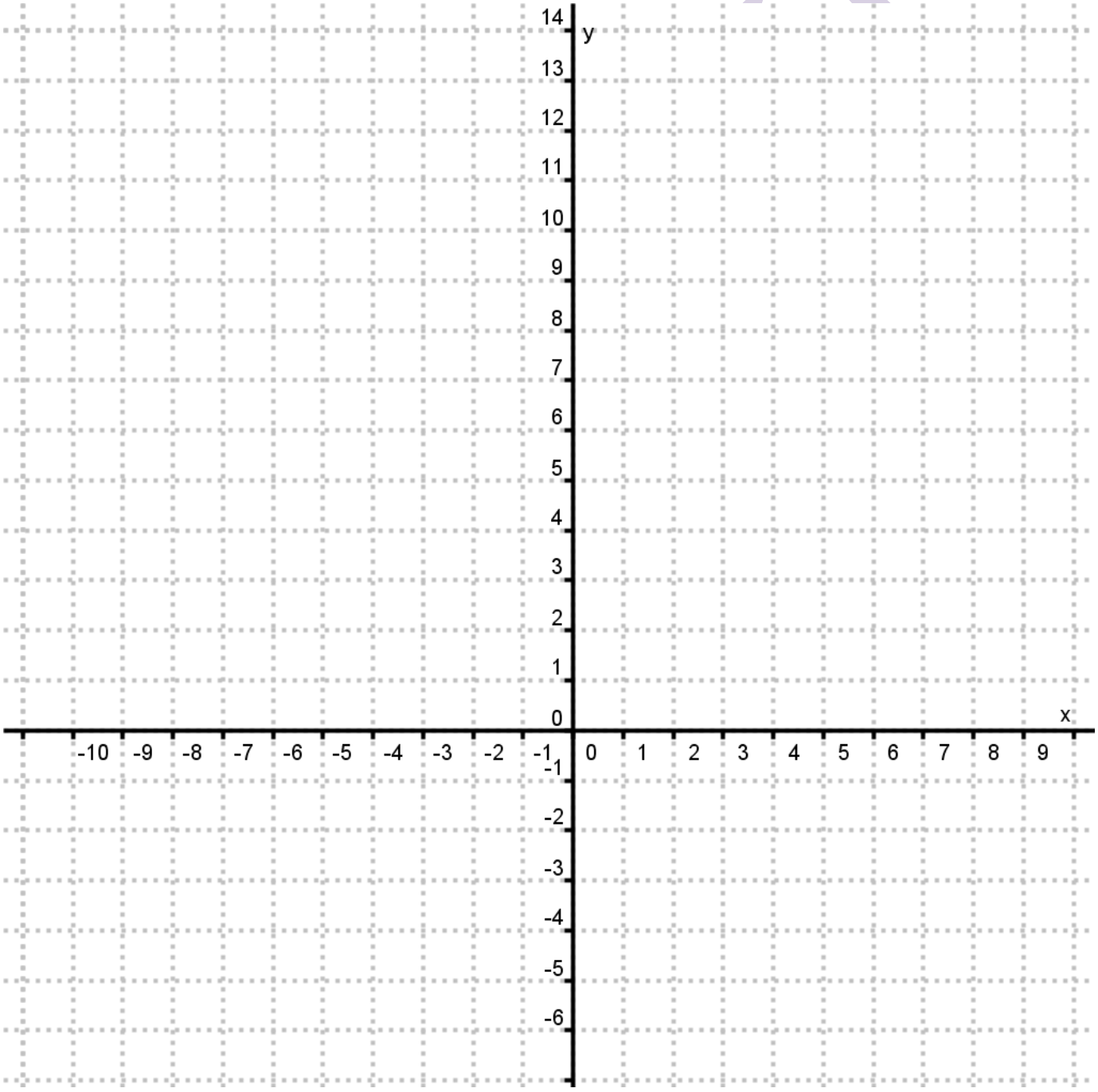
ADDENDUM B

EXAMINATION NUMBER:

CENTRE NUMBER:

QUESTION 4

4.1 4.1.1



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